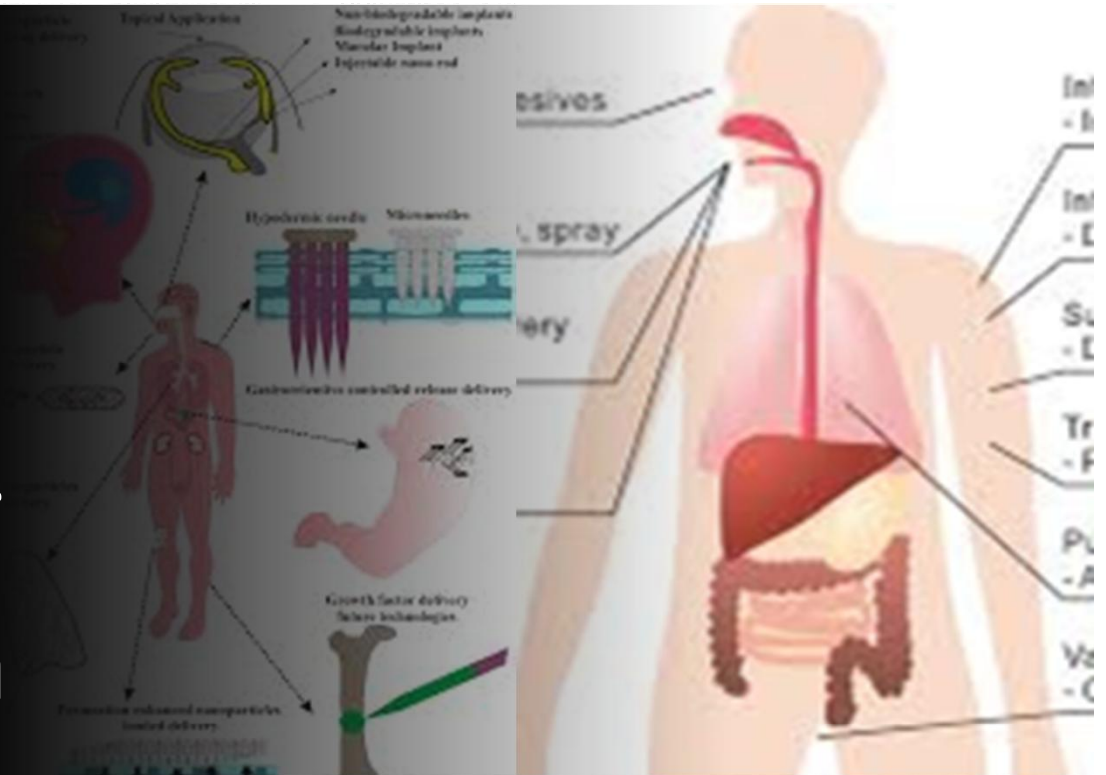


Drug: Route of Administration





Oral administration



Sublingual administration is where the dosage form is placed under the tongue



Buccal administration is where the dosage form is placed between gums and inner lining of the cheek

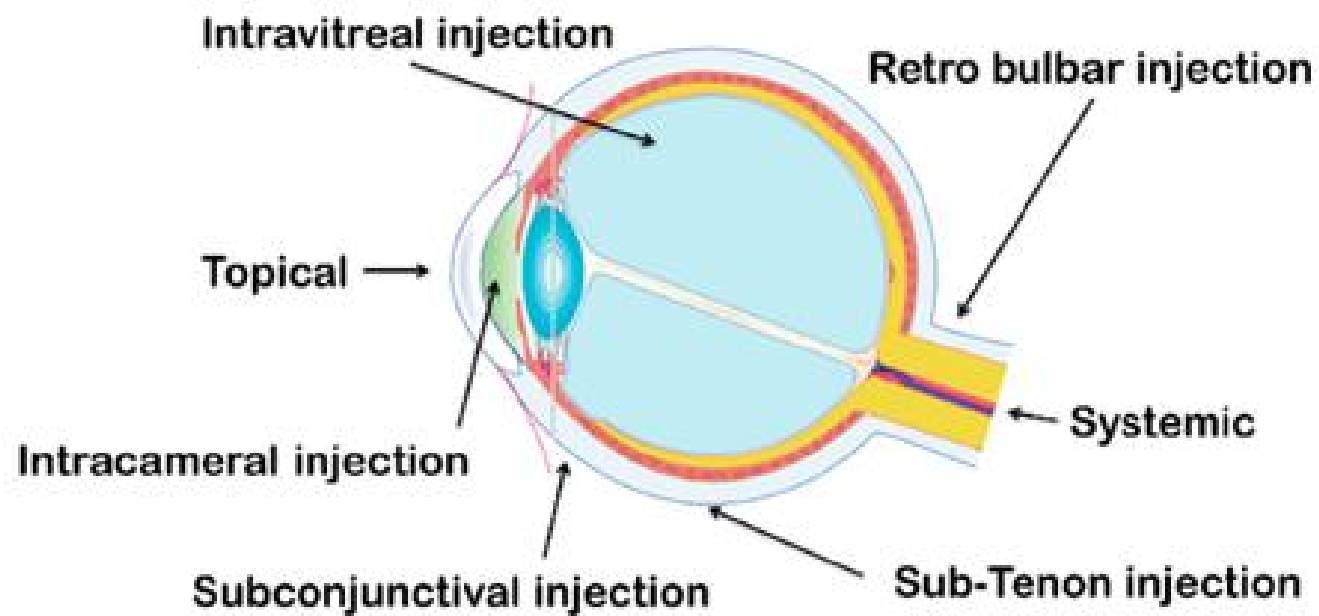
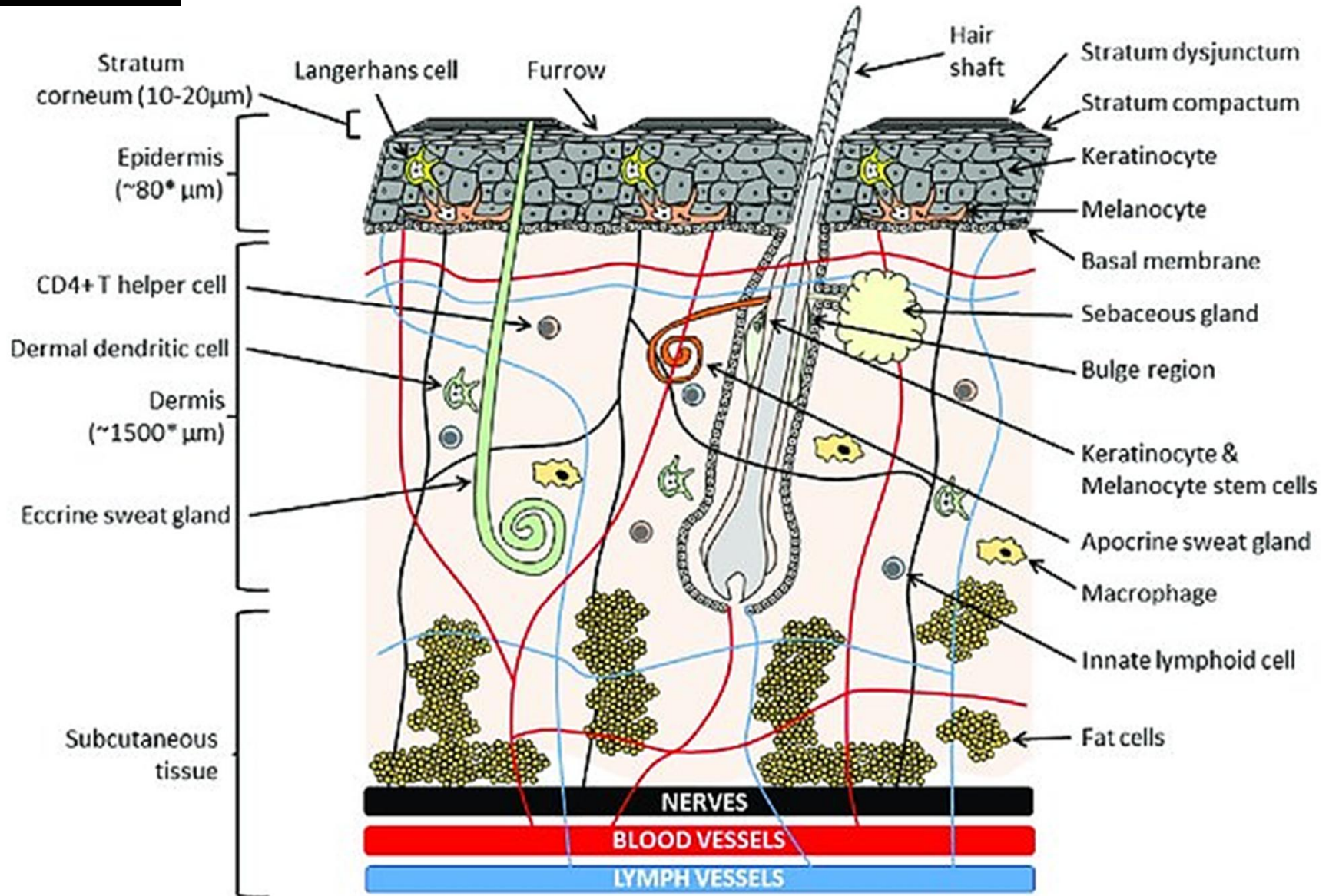


Figure 2. Routes of ocular administration.

Skin in detail

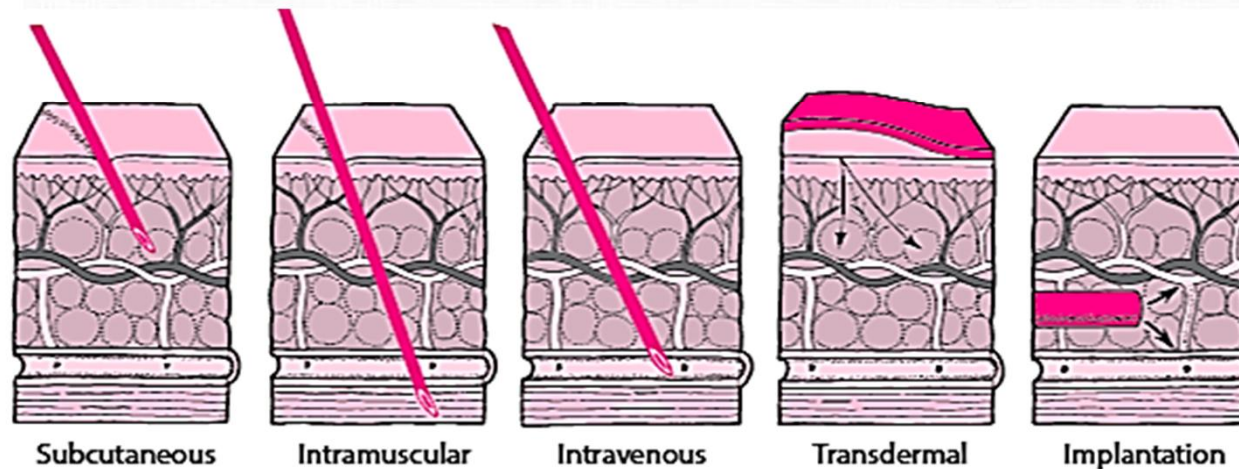
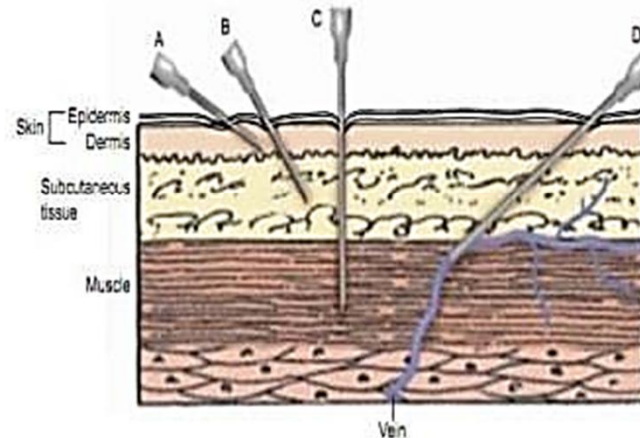


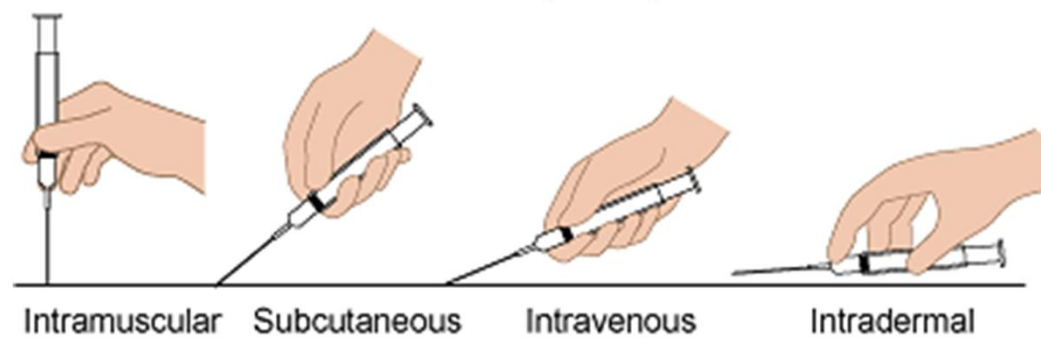
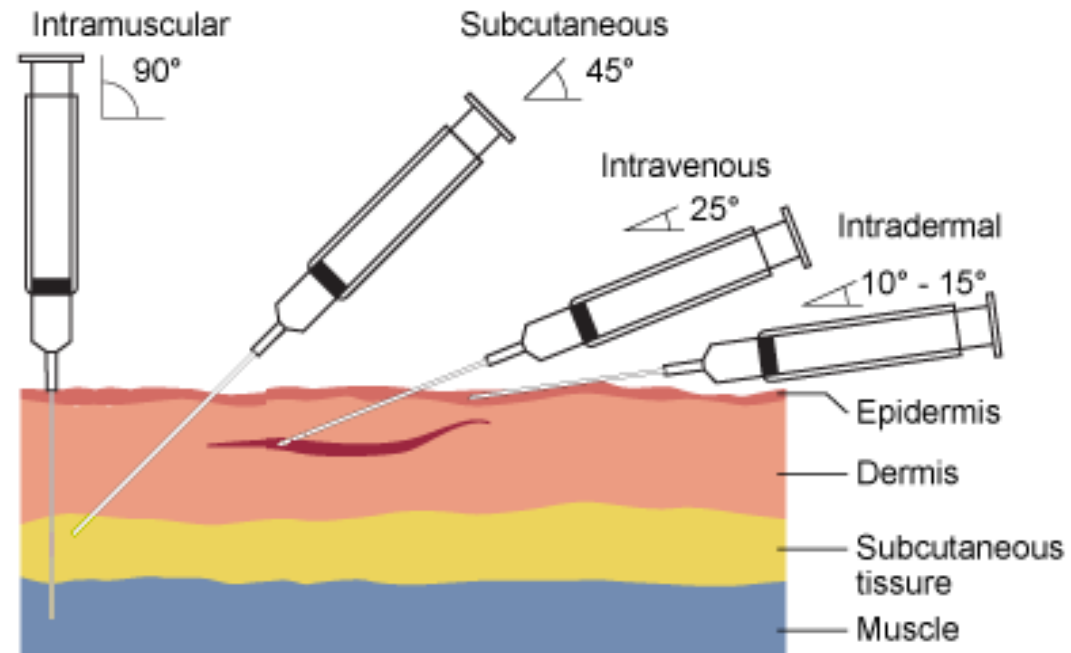
A) Intradermal – inj
into skin

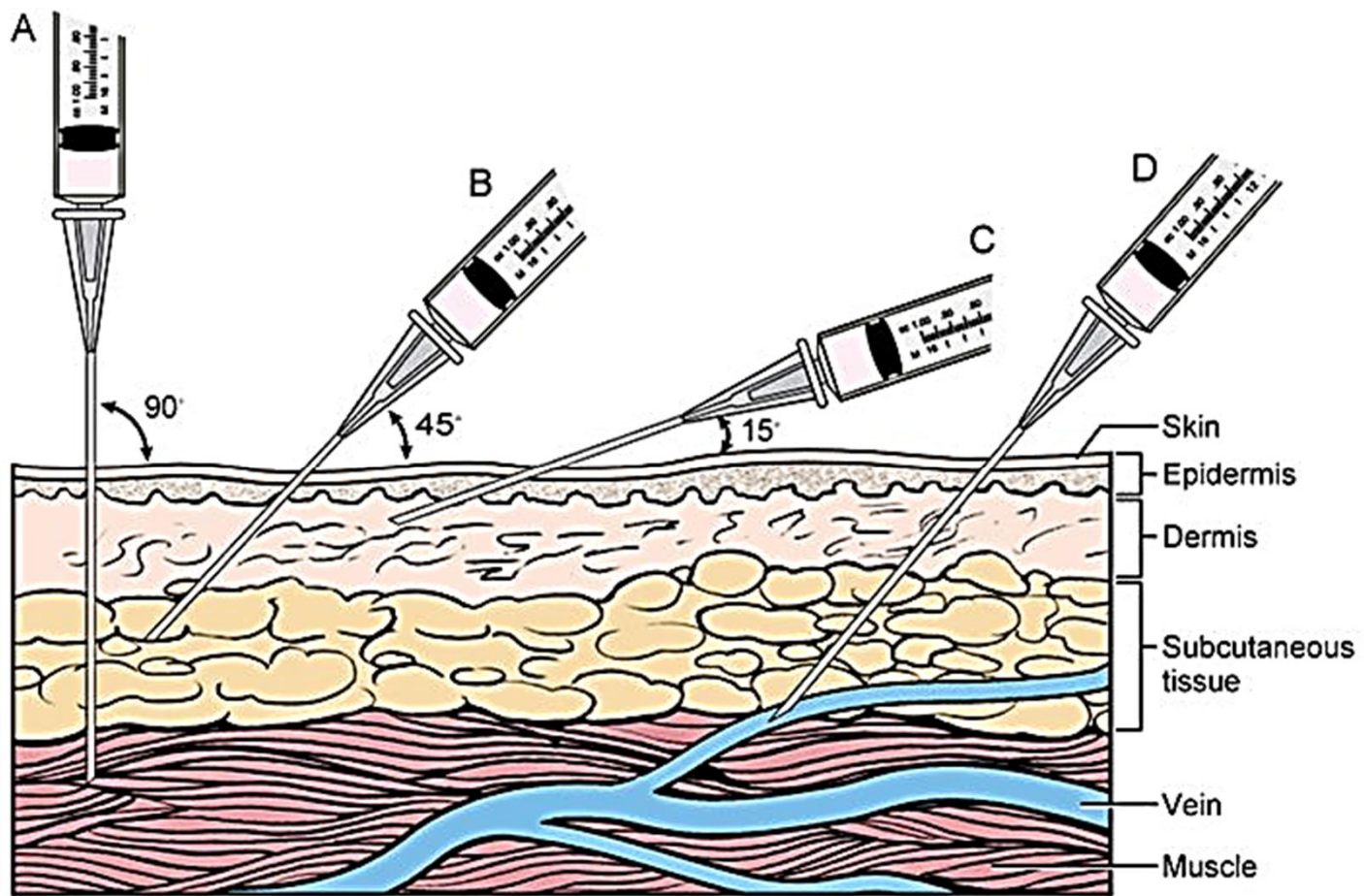
B) Subcutaneous -
Absorption of drugs
from the subcutaneous
tissues

C) Intramuscular (IM)
drug injected into
skeletal muscle

D) Intravascular (IV)-
placing a drug directly
into the blood stream





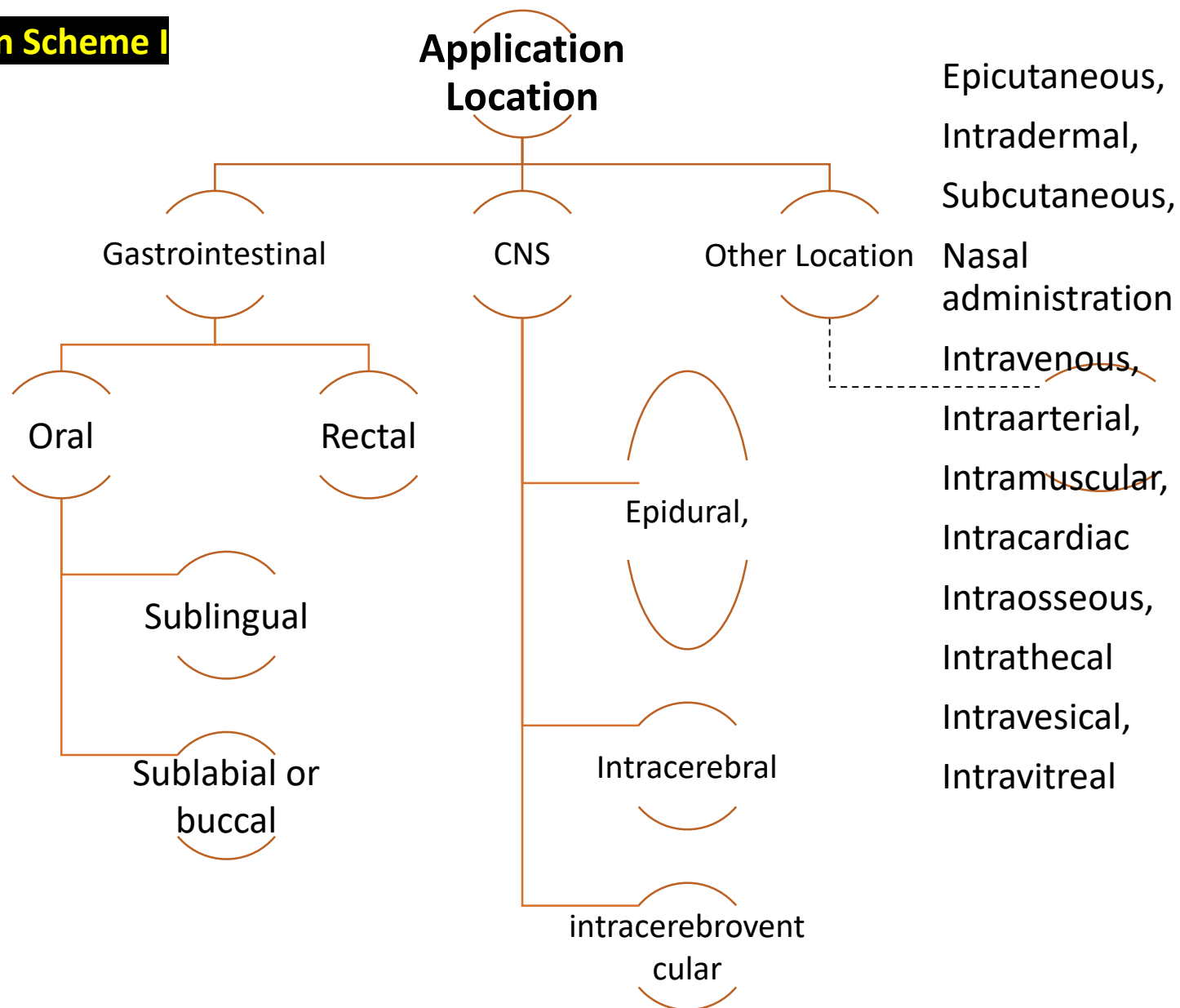


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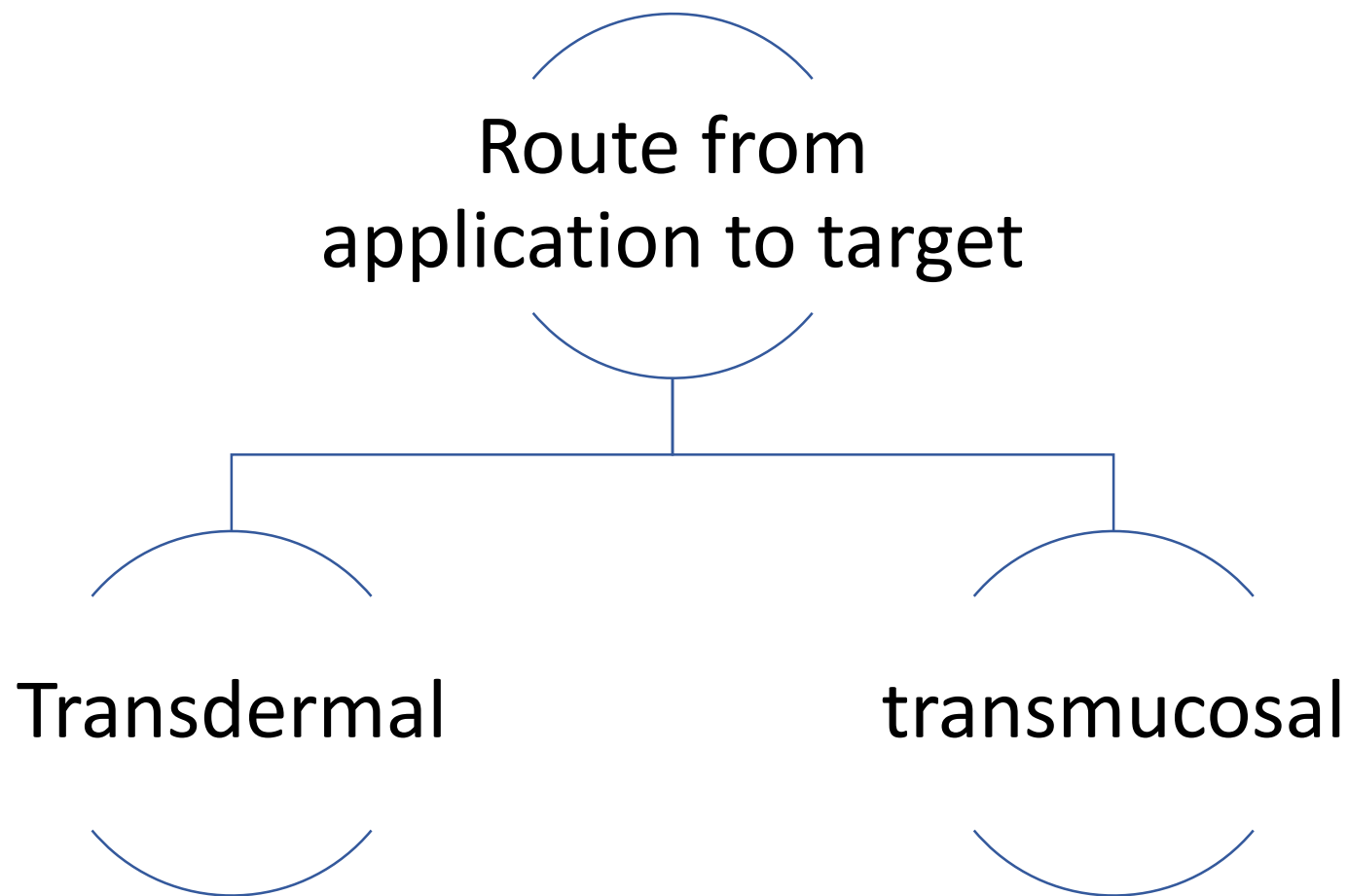
Administration method	Site of injection	Description and use
Epidural Injection	Dura mater of spinal column	Regional pain blockage, injection into epidural space of vertebral spine, often for pain from spinal stenosis and disk herniation
Hypodermic injection	Loose connective tissue and adipose tissue	Advantageous in elderly with less accessibility of veins, for vaccines, self-administration
Intra-abdominal injection/infusion	Abdomen/organ	Direct injection into an abdominal organ such as liver, kidney or bladder
Intra-articular injection	Joint's cavity	Injection of anti-inflammatory drugs into joints
Intra-arterial injection/infusion	Artery	Tissue perfusion with cytotoxic drugs and in vascular imaging procedure
Intracardiac injection	Heart or its musculature	Injection of drugs like adrenaline to arrest a heart attack - now rare, replaced by defibrillation
Intracisternal injection	cerebrospinal fluid of the brain ventricles	Exploration of the cisternal space, for intractable pain of head and neck when all alternate methods have failed

Intracutaneous injection	Within the skin - epidermis (microneedles), dermis (microneedles and needle) and subcutaneous (needle)	Local anaesthesia, desensitisation of allergies
Intradermal injection	Dermis	Typical method used for vaccination and allergy testing
Intra-ocular injection	Eye	Inaccessible infection and severe glaucoma
Intraperitoneal injection/infusion	Peritoneal cavity	Removal of toxins by peritoneal dialysis, especially in renal failure; antibiotics for peritonitis; and chemotherapy of intra-abdominal cancers
Intrapleural injection	Pleural cavity	Local analgesia, local malignancies, pleural empyema and fibrinolysis
Intrathecal injection	Subarachnoid space	Spinal anaesthesia, chemotherapy, pain management and baclofen for spasticity
Intravenous injection/infusion	Directly into the vein	Injection of blood, drug solutions, nutrients, electrolytes. Useful in emergency to get immediate action
Subcutaneous injection/infusion	Loose connective tissue and adipose tissue immediately below the true skin (epidermis)	Advantageous in elderly with less accessible veins, for vaccines, nutrients and various drugs, including by self-administration

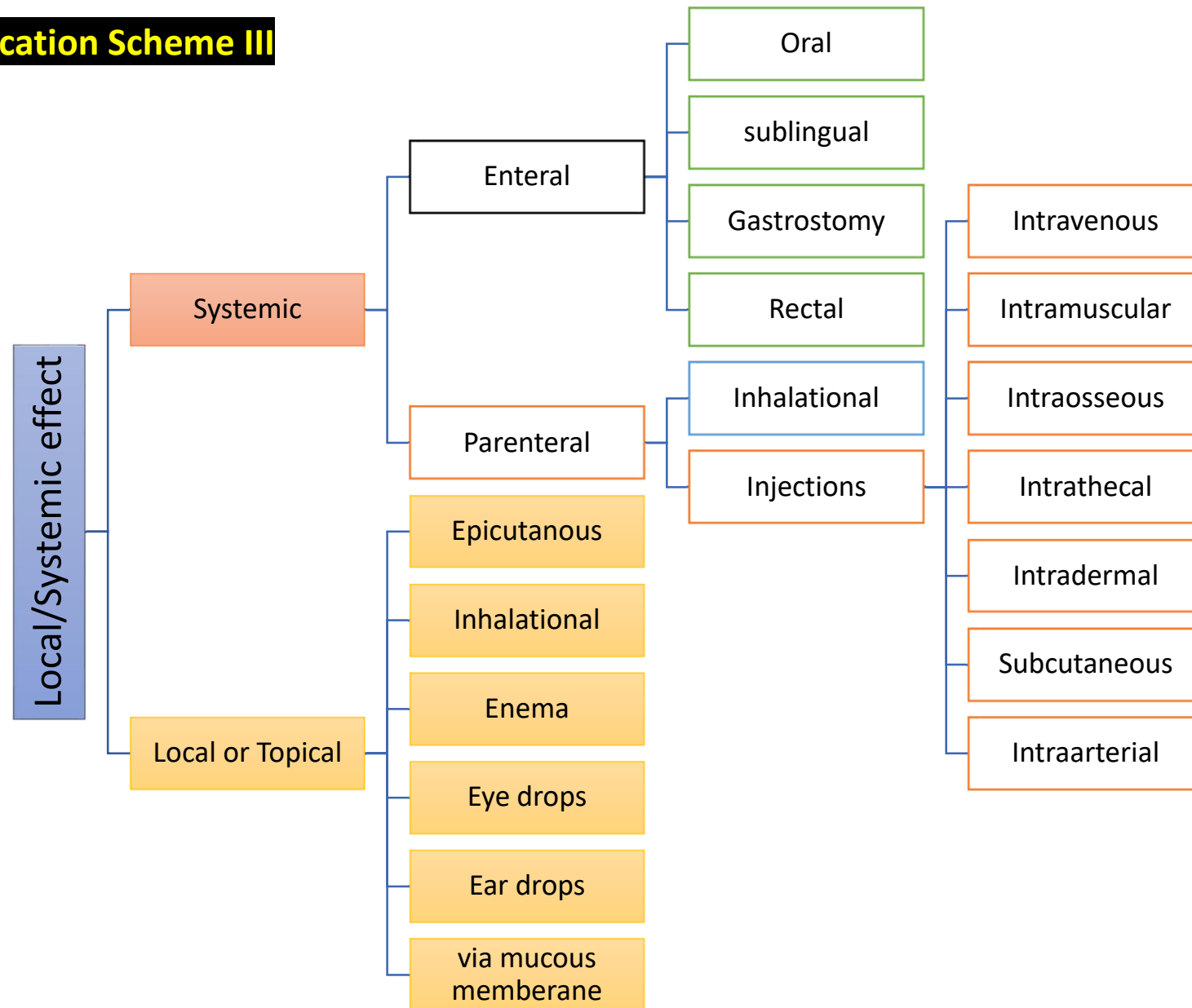
Classification Scheme I



Classification Scheme II



Classification Scheme III



First pass Metabolism

Metabolism of drug in the gut wall or portal circulation before reaching systemic circulation

- ☐ **so the amount reaching system circulation is less than the amount absorbed**

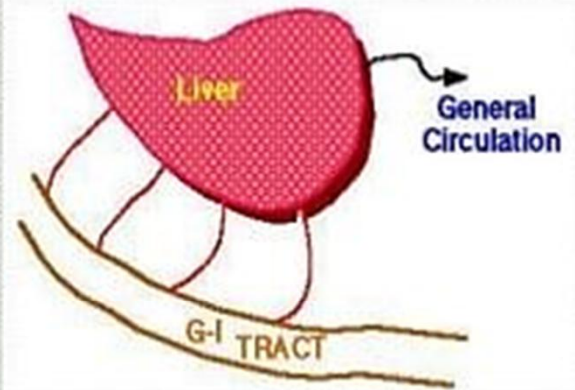
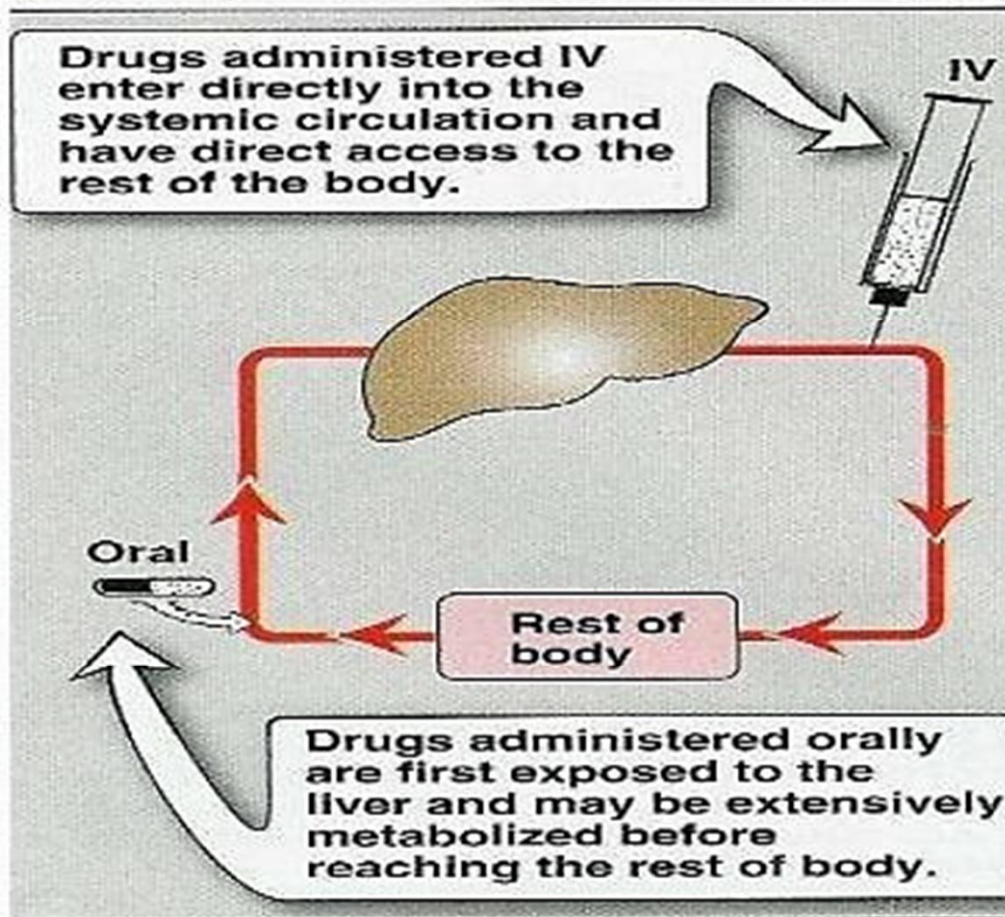
Where ?

- **Liver**
- **Gut wall**
- **Gut Lumen**

Result ?

Low bioavailability.

Short duration of action ($t_{1/2}$).



First-pass metabolism can occur with orally administered drugs.

Route for administration -Time until effect-

☐ intravenous	30-60 seconds
☐ intraosseous	30-60 seconds
☐ endotracheal	2-3 minutes
☐ inhalation	2-3 minutes
☐ sublingual	3-5 minutes
☐ intramuscular	10-20 minutes
☐ subcutaneous	15-30 minutes
☐ rectal	5-30 minutes
☐ ingestion	30-90 minutes
☐ transdermal (topical)	variable (minutes to hours)

PARENTERAL ROUTES

- ❑ Direct delivery of drug in to systemic circulation without intestinal mucosa

Intradermal (**I.D.**) (into skin)

Subcutaneous (**S.C.**) (into subcutaneous tissue)

Intramuscular (**I.M.**) (into skeletal muscle)

Intravenous (**I.V.**) (into veins)

Intra-arterial (**I.A.**) (into arteries)

Intrathecal (**I.T.**) (cerebrospinal fluids)

Intraperitoneal (**I.P.**) (peritoneal cavity)

Intra - articular (Synovial fluids)

